

Published in the

**INDIAN JOURNAL OF ENVIRONMENTAL
PROTECTION**

ISSN 0253-7141

Volume 22, No. 6, June 2002

pp. 707 -716

Joint Forest Management in West Bengal: A Game Theoretic Critique

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Background

In the nineteen eighties and nineties, mostly, the West Bengal Forest Department popularized Joint Forest Management (JFM) with the formation of nearly 3500 Forest Protection Committees (FPCs) for the reserved forests and nearly 90 Eco-Development Committees (EDCs) for protected areas (Government of West Bengal, 1999)¹. This was an outcome of its recognition that conventional forestry, which involves protecting the forest resources from potential offenders with the help of the department's employees, has not worked. In 1991, the state's forest cover as a percentage of total area was abysmally low at 9.03 per cent (Forest Survey of India, 1991). The situation was particularly critical in the southern parts, where most of the forests were heavily or moderately degraded.

In essence, the FPC program encourages the forest-fringe villagers to extract non-timber forest produce (NTFP) in a sustainable fashion, and to stop felling trees for timber.² In return, the incentive being offered is a 25 per cent share of timber felling carried out by the forest department (FD). On the other hand, the villagers living near the *protected* forests are not supposed to extract anything –it is in these villages that the committees are called EDCs. In this (more recent) program the incentives have had to be different - the FD has usually offered various services or products for the economic upliftment of the villages.

Each FPC or EDC is supposed to have one male and one female member from each household in the village/villages that form the FPC. It has an executive body with the local beat officer, who is also the convenor, two local government representatives and six to nine elected villagers as its members. These village committees are supposed to meet at frequent intervals to discuss and solve the various problems associated with protecting the delineated forest area, to make necessary decisions and to allocate responsibilities. Our work covering a cross section of 58 FPCs and over 1000 households in two southern

¹ 'Reserved' forests are 'open-access', where extraction is permitted unless specifically prohibited, whilst 'protected' forests do not allow any kind of extraction unless there are specifications to the contrary.

² Persons other than the locals may also be involved in illegal timber felling. However, this paper implicitly assumes that local connivance is essential for outsiders to carry out their nefarious activities.

districts of West Bengal (Roy, Dutta and Bandyopadhyay, 2001) indicates that the internal functioning of the FPC (more specifically, conflict resolution and the frequency of as well as attendance in meetings) is critical in determining the state of the forest under its supervision.

One cannot deny that this major shift in approach has arrested the appalling rate of degradation, particularly in the southern plains³. But it is equally evident that there is considerable room for improvement. In our view, a game theoretic perspective lends a certain logic to the problem at hand, making it easier for us to locate the necessary measures that would create a strong incentive for forest protection and hence cause a faster rate of afforestation, not only in West Bengal but also in the other Indian states.

Giving the Current Scenario a Game Theoretic Identity

The FD owns the forest areas, and there are state laws specifying how and to what extent they can be used. But because of their hugeness and accessibility vis-à-vis the fringe villagers, and also because of the non-implementation of regulations, they are 'open access' and hence qualify as 'commons'. Hence Hardin's 'tragedy of the commons' applies. That is, extraction occurs where the marginal private gain of each 'herder' (in our case, the villager who uses the forest resources) is equal to the marginal cost of extraction. But there may be a negative externality in the form of resource depletion at non-sustainable levels, and if this cost is not taken into account, the extraction is inefficiently high. As a result, the forests disappear and everyone loses out in the long run. This conclusion is akin to the case of the Prisoners' Dilemma where each prisoner tells on the other, because he assumes that the other will tell on him, that is, he assumes the worst possible action on the part of the other. For our commons, each herder thinks that the others will follow their own private interest, hence he does the same.

³ In 1999 the cover is estimated at 9.4 per cent (Forest Survey of India, 1999).

There are reasons why the commons story may not be one of total defection, but the fact of overexploitation of resources cannot be denied. We shall look into this, and discuss how the problem can be resolved. But first, let us present our particular case in the game theoretic framework.

Let us think of a FD that is omnipotent: the central authority that knows everything and can do whatever it wishes to do. If the FD knows the optimal, sustainable rate of resource extraction and can competently identify and penalize those who extract more than their fair share, it runs the system efficiently and there is no long-term depletion of resources. This is the ‘Leviathan’ solution offered by many for all problems involving negative externalities. The problem with this setup is that the FD may not know the optimal extraction capacity, and it may not be able to identify or penalize the offenders correctly. This has been the case for our forests – in particular, as already mentioned, policing or punishing has been seen to be a very difficult proposition, and this is why the FD has started thinking of other methods of forest preservation. Moreover, the efficient solution of the state control system assumes that there are no costs of such control, but this is not the case.

In spite of the fact that the FD talks about ‘joint’ management, the relationship between the FD and the villagers is far from equal and the FD is still the central authority that dictates the rules. We cannot, thus, talk about the FD as a player in a possible ‘game’. However, what it has done is to introduce elements of a game, by offering payoffs to the villagers and expecting that the consideration of alternative payoffs would induce the villagers to use the forests in a sustainable fashion. Thus the role of the FD has shifted to that of a referee – one who dictates the rules and tries to make sure they are followed, but who allows a game to take place within the boundaries of the rules. We cannot, exactly, call the FD the ‘facilitator’ in the way that the term is used in game theoretic parlance, as a facilitator only contributes the necessary information, encourages a game to take place and helps in conducting the game. In our case the FD is far more powerful. But, we can say that after a point, when it has ensured that the basic guidelines are being followed, it wishes to play the role of facilitator. As a matter of fact it has done this in a very

inefficient and lacklustre fashion, but that is a separate matter that we shall discuss subsequently.

The villagers under an FPC are our players. They are a non-homogeneous group, consisting of the Executive Committee (EC) and other influential persons, the ordinary FPC members, the women (who have no obvious function in the operation of the FPC) and other alienated villagers (who do not take part in any kind of co-operative process but do use the resource). Each individual villager (including the women and alienated villagers) is actually a player as long as he/she extracts forest products, with his/her own independent payoffs and actions. But the relationships between players are not uniform – family members may act as a group, for example; persons of the same ethnic or political group may feel close to each other, the ‘alienated’ persons may act in conjunction and may feel distant from the rest of the community. The EC, an elected body, represents the villagers and the FD does most of its interaction with this body. But this interaction cannot, at present, be called a negotiation. The EC is essentially a mediator between the FD and the villagers, and is thus a partner of the FD in the facilitating process. But members of the EC, like the other villagers, are players. The EC, thus, has two identities.

Hence, as is the case for most real situations, the picture is somewhat complex, with a mix of state impositions at one level and free play at another, and with different roles played by the same individual or organization. We will discuss some of the other forms that JFM could have taken up.

Moreover, JFM involves communication between the actors and is repeated - these are the two features which distinguish the game from the ‘Prisoners’ Dilemma’ outcome which leads to Hardin’s ‘tragedy of the commons’. Thus the players can explore the possibility of co-operation and whether the co-operative outcome is better than the non-cooperative, and they can learn from the outcomes of past actions. And finally, knowledge is imperfect, both on the part of the FD and on the part of the villagers - but this may be corrected to a large extent.

Now, if the level of sustainable extraction of forest resources is known, if an opportunity is given to the players to discuss modes of co-operation so that equal amounts of the resource go to each player and if individual payoffs under co-operation are greater than under non-cooperation, the players will clearly prefer to cooperate. In case payoffs are not fixed and there are uncertainties regarding future payoffs, the historical experience of past payoffs is very crucial in determining the probability of future payoffs. Hence the payoffs or the probable payoffs are very crucial. If these are large enough, the game being discussed here should have a stable equilibrium with all players conforming.

We shall first evaluate the overall structure of JFM, and then concentrate on the payoffs – for as we have seen, it is these payoffs which are crucial to the success of the system.

Weaknesses Inherent in the Existing System

Most importantly, the FD introduced JFM because they realized that centralized control was not possible, yet they retained their position of authority. This position requires, first, that they know the optimal level of extraction. It is not clear that this is known, and it is doubtful whether the FD's felling is based on scientific management. However, we have not investigated this matter – hence let us optimistically assume that the FD is fully knowledgeable regarding the carrying capacity and conducts its felling cycles on that basis. But it is quite clear that for a variety of reasons the FD has no control over the extraction by the villagers. Yet on the one hand it retains the illusion of control and spends a significant quantity of resources on such control. On the other, it admits the impossibility of control and hence asks the FPC to monitor itself. The FD's position, thus, is very unclear.⁴ If it is necessary to retain some control, the areas of control should be well defined and should not overlap the area of operation of the FPC. If a 'game' is a must, the ground rules of the game should be specified and then the game should be allowed to proceed.

⁴ Not only is the FD confused about its role, but also, whatever functions it carries out are often erratic or ad hoc – for example, with regard to punishing offenders, employing temporary labourers or attending meetings.

Secondly, if the creation of the FPC is in order to facilitate a game, the members and the rest of the villagers should operate with as much information as possible which requires much greater interaction amongst themselves as well as support from others – such as the FD, and also those who can give technical information regarding the possible level of extraction and how this limit can be adhered to. In other words, there should be complete transparency and the players should be knowledgeable.

Third, if the objective of the FD is to retain the environment and not to profit from timber sales, the revenues retained by it should be based on what it costs to monitor and facilitate the process of preservation. In particular, it is not clear why the FD should retain 75 per cent of felling returns. Moreover, as long as the villagers feel that the division is grossly unequal, they will not develop the responsibilities that go with rights. It is true that the NTFP returns should also be accounted for, but in a situation where they do not feel that the right of NTFP extraction has been handed over by the FD, there has to be at least an equal division of the timber benefits. Further, if another organization can carry out the operation of monitoring and facilitating at lower costs, the work should be given to such an organization.

Let us, then, look at other possible ways in which the game of JFM can be structured.

Alternative (Game) Scenarios

A. A Two Stage Game

We can think of the process as a two-stage game where, in the initial stage, the players are the FD on the one hand and the villagers on the other. At the second stage, the FD, with the help of the EC, becomes a facilitator in a game where the villagers are the players. In the first stage, then, the villagers are an aggregate body with a single objective. At this stage some essential rules may be imposed (such as the minimum desirable status of the forest) but otherwise the FD and FPC are equal players and they chalk out a strategy that allows both to benefit more in comparison with a non-cooperative strategy. Thus here we assume that both these parties have rights over the forests, as well as responsibilities. Once these rights and responsibilities are chalked out, we enter the second phase of the game where the FD has a certain role as facilitator and monitor (*what* it monitors is well decided in the first round), but the game is now between the villagers. Or, the FD may simply facilitate the process and let the players mutually monitor each other.

The advantage of such an arrangement over the existing one is that the FD and FPC interact at the same level, and the roles are well defined. The initial negotiation ensures a fairer division of returns. Yet the FD is present to supply valuable information and ensure that the environmental needs are not ignored

B. The FD as a Paid Facilitator

An alternative to the above may be the payment of a fee to the FD to cover its costs as a facilitator – this fee is then equally divided amongst the players and gets included as a cost in the payoff structure. This system thus transfers all rights and responsibilities to the people. They get together and arrive at a contract that is acceptable by all. Only after this contract is decided upon, does the facilitating process begin, and the facilitator does nothing that has not already been agreed upon in the contract. This model, if it is to give desirable results, assumes a total understanding of the costs, including the environmental costs.

C. A Least-Cost Facilitator

A system that is more efficient than the above would be to choose the facilitator that provides the service at least cost. This may be an NGO, for example. The choice of the facilitator may also be based on the level of service provided, if we accept that this service need not always be perfect.

D. Privatization

An alternative to all of the above would be to privatize the system and allow market forces to determine the solution. Privatization of the commons, naturally, eliminates its tragedy. It internalizes the negative externality of long-term degradation. However, one problem remains – the private landowner need not be concerned with the larger environmental impacts.

E. Privatization with Legislation / Preservation Incentives.

One way of solving the above problem would be to legislate that the forest cannot be converted to some other use, and has to retain a minimum status. But this would require a degree of supervision. Legislation may be substituted by incentives to preserve, such as tax and other incentives in the private wildlife refuges of Costa Rica (Uphoff and Langholz, 1998).

F. Use Rights

Instead of transferring the ownership totally, the FD may retain the forest but allocate plots of forest land to each villager – the villager enjoys use-rights but with the condition of non-conversion. The user forfeits his rights if he converts. The use rights may or may not be inherited. If inheritance rights are not given, incentives for long term preservation are absent. Hence it would be desirable to grant rights of inheritance. A possible problem with this kind of a situation is that villagers might fell trees from areas owned by others.

All of the above arrangements would yield efficient results, without the overexploitation of the forest resources, if the assumptions hold. The problem, therefore, is to identify those assumptions that may be implemented.

The Payoffs (and Why They Encourage Defection)

The payoff of any strategy would be the difference between the benefits and costs of that strategy. If the villager ‘defects’, which in our case is the illegal felling of trees for timber or the extraction of firewood or fodder or construction materials in a destructive fashion, the benefit-cost structure is given by the following:

Benefits of Defection

1. Returns from the sale of timber
2. Returns from the sale or use of firewood or fodder that is extracted in a non-sustainable fashion
3. Saving the time or labour required to guard the forest

Costs of Defection.

1. Less NTFP
2. No share (25 per cent) from the FD’s felling

3. Less employment opportunities with the FD for plantation or other activities connected with regeneration
4. Cost of time and labour spent in illegal activities
5. The possibility of being jailed or fined for illegal felling or exploitation
6. Social sanctioning of illegal felling
7. Indirect impacts of a worse environment
8. Elephant menace⁵

The costs and benefits of ‘conforming’, that is, not exploiting the forests unsustainably and hence following the regulations set up by the FD, are exactly the reverse, that is, equal to the benefits and costs, respectively, of defecting. Hence, it is important in terms of policy to reduce the payoff of defection (and hence enhance the payoff from conformation) by reducing the benefits and increasing the costs of defection. It is therefore important to study these benefits and costs in greater detail.

Interestingly, the benefits of defection are all material benefits, whereas the costs are material, legal, social and environmental (though the last three may have material implications). It is crucial to note that there is less recognition of the non-material costs in comparison to the obviously material benefits and costs. There may be little knowledge of environmental impacts, perhaps due to the erosion of old ecology-conserving practices and the absence of a new knowledge base that replaces the old. Moreover, social sanctions may be unimportant in the face of abject poverty. We should mention here that the recent literature on managing the commons stresses the great importance of social sanctions at the decentralized level as the factor that has the greatest strength in countering the economic benefits of defection (Uphoff and Langholz, 1998, Sethi and Somanathan, 1996, Ostrom, 1990).

⁵ It is difficult to say whether the elephant menace is a cost of forest degradation – for a certain degree of degradation has caused the elephants to intrude into the villages, but no forests would mean no elephants. Thus in certain areas such as North Bengal villagers often show reluctance to protect the forest because they feel that the forest produces this menace – in other words, they identify a reduction in the elephant menace as a benefit of defection, that is, the increased menace is a cost of conforming.

Secondly, some of the costs may have low effective values. If the legal system is weakly implemented, if offenders are not caught, the probability of being jailed or fined is very low. In a changing social structure, with the intrusion of mass media and the spread of a secular culture, social control systems weaken. They are also weak in heterogeneous communities with mixed ethnic groups. The ‘alienated’ villagers, for example, or even the women who are not included in FPC operations, may be oblivious to sanction.

As regards the material benefits or costs, the monetary returns from illegal timber sale are high because of a high demand for timber. In the vicinity of Jaldapara Sanctuary in North Bengal, for example, there are a huge number of sawmills and furniture shops. Yet here any kind of extraction is illegal and even the FD’s legal felling is very rare. As long as there is a demand, supply becomes inevitable, particularly due to high unemployment and underemployment levels. This benefit from defection, on the other hand, is largely absent in the South Bengal forests because they have little of valuable timber. Firewood extractions are high because there is really no alternative cooking fuel, and to some extent because of population pressures. Fodder extraction, again, depends largely on the cattle population, which is high because of agriculture’s dependence on cattle. Behind all these factors, of course, lies the fact of poverty, which forces the villager to think of even minor gains at the cost of losing one’s respectability.

Item 3 in ‘benefits’ and item 4 in ‘costs’ refer to labour time – both these items are not very important in our context as the opportunity cost of labour is very low given the degree of unemployment and underemployment in the rural sector.

We have seen (Roy, Dutta and Bandyopadhyay, 2001) that NTFP benefits are very significant and act as a strong incentive in forest preservation – thus this is a major cost of defection. On the other hand, we have also seen that legal timber sales by the FD are very infrequent and even when that occurs, the returns to each household is generally low. The erratic and infrequent nature of organized felling means that the villagers would generally assign very low probabilities to obtaining the 25 per cent share of felling in the near future. And they certainly do not have any knowledge of when felling would occur, as the FD

normally does not think it necessary to inform them. Moreover, if future returns are heavily discounted, as is the case for those who live from hand to mouth, the present value of these returns would be even lower. Hence this cost of defection or benefit of cooperation is largely absent, in spite of the fact that this has been publicized as a major incentive for preservation. As regards labour employed for forest-related work, this would also be an important cost of defection if it is regularly done and only conforming FPC members are given work. Although this incentive does exist, it is not very strong.

It is therefore evident that the perceived benefits of defection may be higher compared to the costs, especially in a situation where policing is difficult and laws are lax – even when an offender is apprehended, he is often allowed to go for political or other reasons. The pecuniary benefits of unsustainable felling are backed up by weak social and legal barriers and a lack of awareness of environmental costs, or even costs in terms of lower NTFP yields in the near or not-so-near future.

There are some groups that would have a greater tendency to defect, as their benefits are higher or costs lower. Greater poverty and lower employment opportunities encourage defection. For women, defection may be greater because they have the responsibility of somehow collecting the firewood that would be required for cooking the day's meal. These communities would also not feel the social compulsion of conforming, especially if they are not included in FPC meetings and decision making.

Now, if some of the players defect, there would be a tendency on the part of others who were earlier cooperating to also defect in later periods, on the basis of past experience. This is because they do not obtain the same payoffs that they would have, had everyone conformed, and expectations of future payoffs are even lower. Hence the system becomes unstable. It is therefore essential to ensure that none of the villagers defect by ensuring that the net benefits are more or less equal and reasonably high. We will subsequently discuss the various ways in which this can be ensured.

Policy Implications

A. A Change in the Paradigm

We have looked at the pitfalls of the present structure of JFM. Hence one suggestion is to change this structure in favour of a more clear-cut game, or privatize the system. Privatization seems very improbable under the present circumstances, and the monitoring that has to accompany it may be very difficult. If we choose the first, we can move from the two-stage game to a completely facilitating role of the FD or an NGO. A significantly non-authoritarian role of the FD does assume that the larger environmental interest is served if the villagers take care of their own local interest in preserving the forests. If we do not accept this assumption, we can stick to the two-stage game where the FD dictates some aspects, but its role is better defined and there are no overlapping functions, as is true in the current scenario.

At a more practical level, the FD can clearly dictate certain conditions but should hand over greater rights to the villagers through legislation. If the people are being asked to police themselves, why does the FD go on policing them and that too in a most inefficient manner? It is thus necessary to provide legal identity and status to FPCs by ensuring their registration under the Societies Registration Act (1860) and to involve FPC members in all aspects of decision making with regard to forest protection and management including local microplanning and eco-development schemes. Official delays in registration (as has been observed in many cases) should certainly be eliminated.

Instead of wasting resources on controlling that which it cannot control, it should concentrate on educating the people and strengthening the social norms on the one hand, and finding ways of enhancing their returns from preserving the forests on the other.

One of the more necessary functions of the FD, for example, is to determine the sustainable quantity of extraction of the various NTFP or timber. A question that is often addressed is whether the collection of NTFP degrades the forest or affects its diversity. This aspect should be carefully studied and arrangements arrived at. For example, certain areas of the forest or time periods can be designated for fuel collection or for grazing, and other areas or time periods left alone.

The need of the hour is to increase the efficacy of the FD as a facilitator. Many of the officials at the beat and range levels are indifferent to the whole exercise and are more concerned about the dilution of the policing laws. The concept of being 'equal partners' in forest conservation is bandied about but is by no means practiced. One of the ways in which the FD flaunts its superior position is to conceal as much information as possible from the people.

On the whole, what is truly critical is to scrutinize the present composition of the FD and try to bring about a number of changes. Only those who show some amount of dedication in preserving and regenerating the forests should be taken into service and elaborate training sessions should be conducted so that foresters can operate with full knowledge and efficiency.

Women foresters have shown a great deal of initiative in motivating villagers – moreover, they can interact with the village women and can perhaps convince village heads to allow FPCs with at least 50 per cent women members or perhaps all-women FPCs. Female guards would be able to apprehend female offenders more easily. Thus there should be women at all levels of the FD.

The Executive Committee (EC) is an organ of the FD in its role as a facilitator. In fact, the FD normally interacts with the EC and no one else in the village or FPC. The efficiency of the EC is also, therefore, important – it should be a true representative of the people and at the same time have a strong influence on the people. It should also be able to interact well with the FD. The EC should therefore be carefully chosen. For example, representation in

terms of caste and leadership qualities may not go hand in hand – in which case the choice should be more in the direction of the latter. Moreover, the possibility of training or motivating members of the EC should be explored, so that they can act as motivators and provide critical information to the villagers.

Once the best possible EC is chosen, it is important to ensure that communication between the FPC members is frequent, and for that meetings should be frequent. It is in these meetings that the exchange of information and negotiations take place. Even when there is a lot of disagreement in a meeting and the meeting is inconclusive, the close interaction has positive long-term implications. Once again, we would stress the need to involve all villagers (and especially the women) in the meetings.

The influence of politics in the behaviour of both the FD and the people is evident. One, the ‘bon-o-bhumi sthayee samiti’ of the local panchayat has a representative in the EC, and the panchayat pradhan or his representative is also an EC member. Two, there are other informal political influences. Those belonging to the non-dominant political groups are marginalized. FD officials, also, are bullied by the political bosses who tell them, for example, who should be punished and who allowed to go.

It is not very clear whether the political presence in the EC or FPC always has a negative impact. For, political groups are also representatives of the people and would therefore try to keep the people (as a whole) happy. As Shubik says, the ‘political decision maker must consider interpersonal comparisons of welfare’ (Shubik, 1986). But it is definitely true that the influence of politics complicates the game and this area needs thorough investigation.

B. Improving Payoffs.

Our discussion of the costs and benefits of flouting the norms imply the suggestions that we shall make here. What we will attempt to add here, are ways in which these costs and benefits may be increased and decreased, respectively.

Women and other alienated groups have to be involved in the whole act of preservation. Unless all villagers, male and female, are in an equal position in terms of both rights and responsibilities, the game cannot yield a cooperative solution. Thus women, most importantly, have to have a major role in the FPCs, and perhaps an even greater role than the men, as *they* are usually the 'herders'. The alienation of women is perhaps the single most important defect of JFM. Unless all herders have significant net payoffs, no attempts to increase these payoffs would work

We will not delve into all the possible ways in which women can be involved. But some of the obvious methods are to increase the female staff in the FD, to involve NGOs dealing with women, to try to change the attitudes of the husbands/fathers, to fix meeting timings so that women can attend and to give women direct access to some of the benefits. .

Similarly, ensuring equity in benefit sharing is an essential prerequisite for building up confidence amongst the socio-economically-backward sections. For example, benefits in ecotourism (e.g. the supply of food and other amenities to forest lodges) should be distributed uniformly.

The question of equity also arises in terms of the forest to household ratio. The FD presently assigns the forest patches to adjacent communities without any scientific delineation criteria based on the sustainable density holding capacity. Therefore the above ratio varies significantly for different FPCs leading to a greater possibility of unsustainable extraction levels in some areas.

A critical measure would be to reduce the returns from and increase the difficulty of illegal felling. This would require the elimination of the channels through which such illegal timber is sold. One obvious move would be to close the sawmills or furniture shops and to apprehend the timber Mafia. The fines of illegal acts should be prohibitive and jailing should be a common punishment. Political intervention is one important reason why these activities continue. One approach would be to remove the political influence, another would be to educate the powerful groups in an area and involve them in conservation.

The FPC can also impose the fines. Alternatively, the FPC may not allow the offender to take his/her share of the 25 per cent felling returns or may bar him/her from entering the forest to collect NTFP for a certain period of time.

To eliminate the over-extraction of firewood or fodder, it would first be necessary to know what the maximum levels of extraction should be, and design the behaviour of the villagers accordingly. But if such over-extraction is absolutely necessary, it will go on. Thus it would be important to try to develop alternative sources of fuelwood and fodder. Villagers can be encouraged to grow supplies in their own homestead or agricultural land or village commons. Some of the forest land can also be used to grow fodder plants. When planting trees, attention should be given to the needs (mainly fuelwood and fodder) of the people.

Substitutes of wood as fuel (e.g. CNG or kerosene) or of cattle (e.g. tractors) would help – but neither is very feasible in the present scenario. There have been attempts to introduce stall feeding cattle (Jersey cows) but they have only turned out to be expensive propositions. However, economizing on fuel use is quite possible (such as with the use of smokeless *chullahs*).

Enhancing employment opportunities would reduce the number of people taking on the risk of wood theft. Opportunities connected to a certain forest area should be provided to members of the FPC that takes care of that area, so that the link between preservation and its ensuing benefit is obvious. Ecotourism is one such area that is developing but is still largely unexplored. Other than employment, various kinds of support may be given to the village to improve its agricultural productivity. If the FD does spend money on the development of the village, it should not be doled out individually (as has happened sometimes in the past) but should go towards developing the village as a whole. Part of the payoffs received by FPC members can also be used for the creation and maintenance of community assets. Hence irrigation facilities may be provided, or there may be other infrastructural improvements. It is more important, from a long run perspective, to provide technical assistance in the areas of earthen dams, irrigation engineering, agroforestry,

pisciculture, roads etc. Technical assistance would also be valuable in handling the elephant menace. We would like to stress that all these developmental activities are related in some way to forest preservation. Care should be taken to not introduce anything that improves the economic condition of the village but has a negative impact, however indirect, on preservation.

In this context, it may be mentioned that in villages adjacent to the *protected* forests, 'micro-planning' has been introduced as a benefit of forest preservation by the EDCs. The project is both difficult and critical as the EDCs are not supposed to receive any other benefits (such as from NTFP). The program is in its nascent stage; we will only suggest that such micro-planning should involve other government departments and NGOs for training and implementation of the projects and should take the needs of the villagers into account. The factors that should be kept in mind are long run sustainability, indigenous technology and equity in benefit disbursement. For example, the distribution of *Supari* saplings has been one of the more successful measures, whilst there have been many expenditures such as on farm animals or handlooms which have gone down the drain.

To enhance the returns from NTFP, it is essential to plan the species mix with the needs of the villagers in mind. In combination, processing units for marketable NTFP can be set up for each FPC in easily accessible places and the marketing of the goods arranged for. There are numerous ways in which this market can be improved, yielding lucrative occupations for many villagers. LAMP⁶, a tribal co-operative, does purchase and market some items but the purchase prices are very low and it generally does not look into processing. Moreover, LAMP appears to be in a moribund state. But other NGOs or government departments can be involved in processing and marketing activities.

Whilst NTFP collection is not allowed in the protected forests, the villagers extract whatever NTFP they need while the FD looks the other way. This is inevitable, as there are no feasible/sufficient alternatives in terms of fuel or fodder. The flouting of laws has such a

⁶ Large Scale and Multipurpose Cooperative Society

negative impact that it is worse than a scenario where the laws are absent. Hence the FD should seriously consider allowing the use of certain NTFP, with, of course, more stringent restrictions for those which are currently protected. For example, certain periods in the year or certain patches of the forest may be demarcated for NTFP collection.

The (25 per cent) share that the FPC members get from the FD's felling operations is too low – if for no other reason, then because there is a symbolic implication that the villagers are 'lesser amongst equals'. But even if they had got this lesser share, it would not have been so bad. The fact is, as we have seen, that there are no systematic sales and the returns are very small. Very few FPCs have experienced a sale and got their percentage, which has amounted to very little money. The amount and probability of a payoff are thus both low – resulting in an expected payoff that is close to zero.

This state of affairs should be totally altered. Ideally each FPC should have a felling each and every year, so that the payoffs are regular. The selection of the patch to be felled should be based on its age and other factors. The total forest area can be organized in such a way as to allow for such yearly felling. Once this is ensured, the question of raising the share to 50 per cent or more arises. One way of deciding what the share should be is to calculate the costs of a rationalized FD and determine the FD's share.

There is a great need to educate and inform the people regarding the present status of the forest, what the ideal status should be, why this is the ideal (that is, discuss the environmental implications), the need to reach and maintain the ideal, what has to be done to stick to the ideal and various technical know-how related to this. In other words, there should be a holistic environmental education. This can be done by the FD, or by NGOs who have the skill to communicate and impart education. Although traditional communities had their own systems of resource preservation, these systems have eroded and they need to be replaced by new systems. These educational sessions can be held during FPC meetings.

Finally, social sanctioning, which already exists but perhaps at a mild level, should be seen as the greatest priority in present forest policy. This, too, can be strengthened through education and by involving all extractors in JFM. The poorest groups whose desperate needs force them to ignore such sanctioning can be targeted specially and provided with alternative economic means. The FPC and its EC have a major role to play in such sanctioning. Enhancing the powers of this body and ensuring that this body meets frequently would therefore contribute towards a greater role played by social disapproval.

Summary

Our game theoretic view of Joint Forest Management in West Bengal indicates that our policy makers should redefine and bring clarity to the role of the forest department as a facilitator and increase its efficiency in this role, as well as introduce equity in the sharing of both rights and responsibilities: between men and women and between socio-economic groups. A crucial step in the right direction would be to increase the payoffs to forest protection by (amongst other things) enhancing the returns from non-timber forest produce (NTFP) and regularizing as well as enhancing felling returns. At the same time, the state should look at the factors that are connected to illegal timber felling or the over-extraction of NTFP and try to eliminate those which encourage such behaviour as well as enhance those which discourage such behaviour. Steps have to be taken to improve the operation of the FPC - in particular, ensure frequent meetings for the resolution of conflicts as well as the dissemination of knowledge. While doing this, one has to be conscious of the political factor and take this into account as a motivating factor in whatever strategy each FPC member or the Executive Committee adopts. Finally, methods have to be evolved to strengthen the role of social sanction as a strong disincentive for potential offenders.

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